An Attention-Guided Neck Feature Fusion Framework for Drone Detection Using YOLOv8 and CBAM

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***Abstract*—Small unmanned aerial vehicles (UAVs) are difficult to detect in outdoor images when they are distant, and they are easily confused with birds, vegetation, clouds, and buildings. This paper proposes an attention-guided enhancement of the You Only Look Once (YOLO) detector YOLOv8x by incorporating the Convolutional Block Attention Module (CBAM) at two selected feature-fusion stages in the path aggregation neck. Specifically, channel attention followed by spatial attention is introduced after the 40×40 and 80×80 feature-fusion stages. This refines the multi-scale features before the detection head while leaving the backbone unchanged. On the supplied validation split, the model reaches 92.93% mAP@0.5 and 57.10% mAP@0.5:0.95, gains of 3.71 and 3.34 percentage points over the baseline. The two modules add 64,198 parameters (+0.094%) and 0.0018 GFLOPs (+0.0007%). Furthermore, the ablation study shows complementary contributions from channel and spatial attention. A perceptual-hash audit shows that 9.5% of the validation images have near-duplicates in the training set, so the results are limited to the supplied validation split and do not establish small-object-specific gains or generalization to independent datasets.**

***Index Terms*—drone detection, UAV detection, YOLOv8, CBAM, attention mechanism, object detection, feature fusion**

I. Introduction

Unmanned aerial vehicles (UAVs), or drones, are now used well beyond military settings, in tasks such as infrastructure inspection, precision agriculture, package delivery, mapping, disaster response, and surveillance. Their low cost and agility also raise security concerns such as airspace intrusion, smuggling, covert surveillance, and collision risks near airports, as part of a wider rise of small autonomous robots [1].

Spotting low-flying drones in cluttered outdoor scenes is still difficult. Seen from a ground camera, a drone may cover only 16×16 to 32×32 pixels of a high-definition frame, so fine details such as propellers, motor mounts, and the airframe are largely lost after downsampling. Its appearance also changes with viewing angle, lighting, motion blur, and occlusion, while clouds, birds, and foliage can produce drone-like patterns.

Preserving information across scales is therefore essential. Feature Pyramid Networks (FPN) [2] and Path Aggregation Networks (PANet) [3] merge features from different resolutions, but this fusion can also carry background noise into the feature maps and blur the edges of small objects. Attention mechanisms address this by weighting features according to their channel and spatial importance, and the Convolutional Block Attention Module (CBAM) [4] does so by applying channel attention followed by spatial attention.

In this work, two CBAM modules are added to the path aggregation neck of the You Only Look Once (YOLO) detector YOLOv8x, after the 40×40 fusion stage at Layer 12 and the 80×80 fusion stage at Layer 15. These are the points where features from different scales meet, so refining them here helps suppress background clutter and keep the visual cues of small drones before detection.

The main contributions of this work are as follows:

- We evaluate YOLOv8-CBAM, an attention-guided architecture intended to refine multi-scale features in visually complex outdoor environments; the available experiments do not isolate small-object-specific performance.
- We introduce a targeted neck-level attention strategy by placing two CBAM modules after Layers 12 and 15. The YOLOv8x backbone is kept unchanged.
- We conduct an ablation study by comparing the baseline YOLOv8x with channel-only attention, spatial-only attention, and the complete CBAM configuration.
- On the supplied train/validation split, the proposed framework improves mAP@0.5 and mAP@0.5:0.95 by 3.71 and 3.34 percentage points, respectively. The measured overhead is 64,198 parameters and 0.0018 GFLOPs, and we audit the dataset for train/validation near-duplicates.

II. Related Work

A. Deep Learning Object Detection

Object detectors built on deep learning are usually grouped into two-stage and single-stage designs. Two-stage methods such as Faster R-CNN [5] propose and then refine candidate regions, which is accurate but slow, whereas single-stage methods such as SSD [6] and the YOLO family predict boxes in a single pass and suit real-time use. YOLOv4 [7] brought in a Cross Stage Partial Network (CSPNet) backbone, PANet

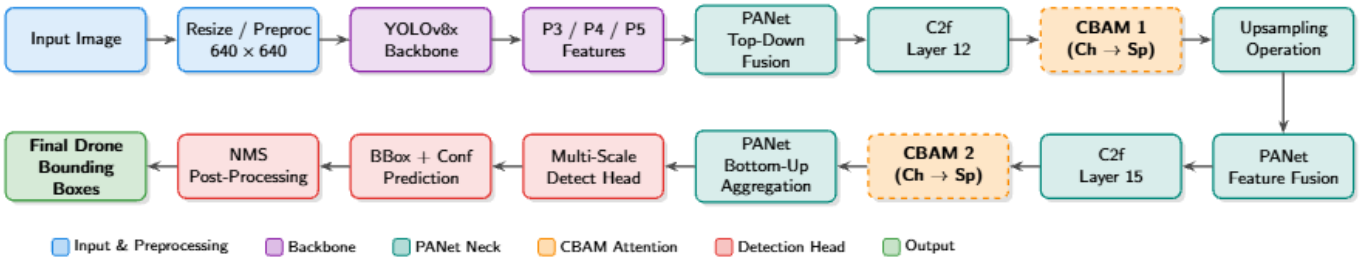


Fig. 1. End-to-end pipeline of the proposed YOLOv8-CBAM drone detection framework.

feature routing, and mosaic augmentation, while YOLOv7 [8] improved efficiency through extended efficient layer aggregation. YOLOv8 [9] then added the C2f (Cross Stage Partial with two convolutions) block for better gradient flow and feature reuse, together with an anchor-free decoupled head.

B. Multi-Scale Representation and Attention

Small objects are hard to detect because deeper layers gain semantic information at the cost of spatial resolution. FPN [2] handles this with a top-down path and lateral connections that merge features across scales, and PANet [3] adds a bottom-up path so that low-level spatial detail reaches the deeper layers more easily. Improved YOLOv5 variants have applied similar ideas to general object detection [10] and to finding small defects on cluttered printed circuit boards [11].

Attention modules take a different route by weighting features according to how useful they are. Squeeze-and-Excitation (SE) networks [12] learn channel weights from globally pooled features, while CBAM [4] adds a spatial branch so that the network learns both which channels and which locations matter. Coordinate Attention [13] builds position information into channel attention by pooling along each spatial direction, and CBAM_SAUNet [14] showed that combining channel and spatial attention helps with difficult corner cases in segmentation.

III. Dataset and Problem Formulation

A. Dataset Attributes and Annotation

This study uses the publicly available YOLO Drone Detection Dataset from Kaggle [15], which contains 1,359 images of a single class, drone, captured against clear and overcast skies, urban skylines, and dense vegetation. The images are split into 1,012 for training (74.5%) and 347 for validation (25.5%). The images come from web photographs and from frames of several videos, identifiable by their file names. A 64-bit perceptual-hash (pHash) audit of all train–validation pairs shows that 33 of the 347 validation images (9.5%) have a near-duplicate (Hamming distance ≤ 4) in the training set, 30 of them identical pictures stored under different names, and 48 (13.8%) have a same-scene match (≤ 10). The supplied split is therefore not source-independent. We also built a scene-grouped train/validation/test split (975/197/187 images, no cross-split pair within 10 bits) for future evaluation; all

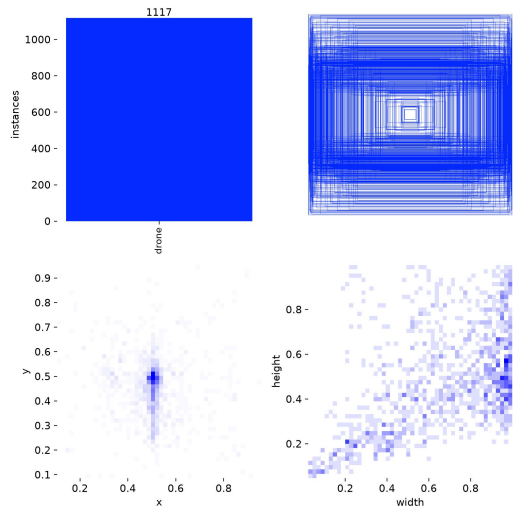


Fig. 2. Statistics of the training annotations: (a) number of drone instances, (b) overlaid bounding-box shapes, (c) distribution of normalized box centers, and (d) joint distribution of normalized box width and height.

results below use the supplied split. Each drone is labeled with a normalized bounding box:

$$\mathbf{b} = (x_c,\, y_c,\, w,\, h) \tag{1}$$

where (x_c, y_c) is the box center and w and h are its width and height. All images are letterbox-resized to 640×640 pixels so that their aspect ratio is preserved.

As Fig. 2 shows, the training split holds 1,117 drone instances, slightly more than one per image. Box centers cluster near the middle of the frame. Measured after letterboxing to 640 pixels with the COCO thresholds, only 13 of the 1,117 training boxes (1.2%) and 25 of the 369 validation boxes (6.8%) are small ($< 32^2$ pixels), while 81.8% of the validation boxes are large ($\geq 96^2$); the benchmark is dominated by close-range drones.

B. Formal Detection Objective

The detector is trained by minimizing a combined loss function:

$$L_{total} = \lambda_{box} L_{CIoU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL} \tag{2}$$

Here, Complete Intersection over Union (CIoU) loss is used for bounding-box localization, Binary Cross-Entropy (BCE)

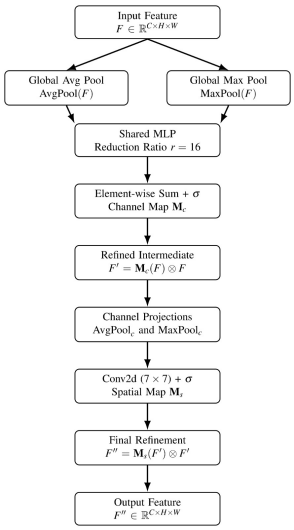


Fig. 3. CBAM sequential channel-then-spatial attention workflow.

loss handles classification, and Distribution Focal Loss (DFL) helps improve the localization of small targets.

IV. Proposed YOLOv8-CBAM Framework

A. Overall Architectural Pipeline

As shown in Fig. 1, YOLOv8-CBAM keeps the standard YOLOv8x backbone and decoupled anchor-free detection head, and changes only the path aggregation neck. Two CBAM modules are placed after Layer 12 and Layer 15, where the 40×40 and 80×80 features are fused, so that the combined features are refined before they pass through the bottom-up pathway to the detection head and non-maximum suppression.

B. Backbone and Multi-Scale Feature Hierarchy

YOLOv8x is used as the base network in the proposed framework. Its C2f-based backbone downsamples the input by factors of 8, 16, and 32, producing three feature pyramid stages:

- P3 (80×80, 320 channels): high-resolution features with fine spatial details.
- P4 (40×40, 640 channels): medium-resolution features balancing detail and context.
- P5 (20×20, 640 channels): deep features with wide receptive fields from the Spatial Pyramid Pooling-Fast (SPPF) module.

C. CBAM Formulation

The internal operation of the CBAM module is illustrated in Fig. 3. For an intermediate feature tensor $F \in \mathbb{R}^{C\times H\times W}$, CBAM applies two attention mechanisms sequentially: channel attention followed by spatial attention. The channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$ is obtained by applying average and max pooling across the spatial dimensions. These pooled features are passed through a shared multilayer perceptron (MLP) with a reduction ratio of $r = 16$, and the results are combined and passed through a sigmoid function:

$$M_c(F) = \sigma\big(W_1\big(\delta\big(W_0\big(F^c_{avg}\big)\big)\big) + W_1(\delta(W_0(F^c_{max})))\big) \quad (3)$$

where W_0 and W_1 represent the shared MLP weights and δ denotes the ReLU activation. The resulting channel attention is then applied to the input feature map through element-wise multiplication:

$$F' = M_c(F) \otimes F \quad (4)$$

Next, the spatial attention module applies average and max pooling along the channel dimension, combines the resulting spatial descriptors, and processes them using a 7×7 convolution followed by a sigmoid function:

$$M_s(F') = \sigma\big(f^{7\times 7}([\mathrm{AvgPool}_c(F');\, \mathrm{MaxPool}_c(F')])\big) \quad (5)$$

Finally, the spatial attention map is applied to the refined feature map to obtain the final CBAM output:

$$F'' = M_s(F') \otimes F' \quad (6)$$

D. Targeted Attention Placement

Instead of adding attention modules throughout the entire network, we selectively place two CBAM modules at important feature-fusion points in the neck:

- CBAM-1 (Layer 12): This module is placed after the C2f Layer 12 and operates on a $640 \times 40 \times 40$ feature map. At this stage, features from the upsampled P5 layer are combined with the P4 features.
- CBAM-2 (Layer 15): The second CBAM module is placed after C2f Layer 15 and operates on a $320\times 80\times 80$ feature map. Here, the top-down semantic information is combined with the high-resolution P3 features. Attention here is intended to preserve fine spatial detail before the bottom-up aggregation stage.

E. Attention Integration

The attention modules are attached by module wrapping, without changing the YOLOv8 layer definitions: the C2f modules at indices 12 and 15 are replaced by a wrapper that runs the original block and then CBAM on its $[B, 640, 40, 40]$ and $[B, 320, 80, 80]$ output, where B is batch size (Table I). The CBAM output is blended as $Y = F + \gamma(F'' - F)$ with a learnable scalar γ initialized to zero, so training starts from the exact pretrained network. The wrapper is inserted after the pretrained YOLOv8x weights are loaded, and these weights are fine-tuned jointly with CBAM, not frozen (only the fixed DFL projection is not trained). All CBAM tensors, including γ, are registered in the optimizer, are stored in and restored from Ultralytics checkpoints, and the wrapped model exports to ONNX (opset 17) with a maximum deviation of 9.7×10^{-4} from PyTorch. The wrapping is applied inside the trainer, because stock Ultralytics rebuilds the network at the start of training.

TABLE I
SELECTED BACKBONE AND NECK FEATURE STAGES OF YOLOV8X

Layer Index	Module Type	Channels	Resolution	Operational Role
0	Conv2d ($k = 3$, $s = 2$)	80	320×320	Low-level stem
1	Conv2d ($k = 3$, $s = 2$)	160	160×160	Downsampling
2	C2f ($n = 3$)	160	160×160	Feature extraction
3	Conv2d ($k = 3$, $s = 2$)	320	80×80	Downsampling
4	C2f ($n = 6$)	320	80×80	P3 multi-scale feature
5	Conv2d ($k = 3$, $s = 2$)	640	40×40	Downsampling
6	C2f ($n = 6$)	640	40×40	P4 multi-scale feature
7	Conv2d ($k = 3$, $s = 2$)	640	20×20	Downsampling
8	C2f ($n = 3$)	640	20×20	Deep context extraction
9	SPPF ($k = 5$)	640	20×20	P5 multi-scale feature
12	C2f	640	40×40	Neck fusion / CBAM-1
15	C2f	320	80×80	Neck fusion / CBAM-2
22	Detect Head	Multi	Multi	Decoupled detection

TABLE II
HYPERPARAMETER AND TRAINING SPECIFICATIONS

Hyperparameter	Experimental Setting
Base Model Architecture	YOLOv8x
Initialization	Pretrained Checkpoint
Input Image Dimension	640×640 (Letterbox Padded)
Total Training Epochs	32
Mini-Batch Size	16
Optimization Algorithm	SGD with Momentum (0.937)*
Weight Decay Penalty	0.0005
Initial Learning Rate (η_0)	0.01*
Channel Reduction Ratio (r)	16
Spatial Convolution Kernel	7×7
Target Foreground Classes	1 (drone)
Training Split Partition	1,012 Images
Validation Split Partition	347 Images

*M2–M4. M1 used `optimizer=auto` (AdamW, $\eta_0 = 0.002$).

V. IMPLEMENTATION AND TRAINING SETUP

A. Training Protocol and Hyperparameters

The experiments were conducted using PyTorch with automated mixed-precision support, with the YOLOv8x backbone initialized using pretrained weights. The main experimental settings are summarized in Table II. All configurations were trained for 32 epochs, and the checkpoint with the best Ultralytics fitness ($0.1\,\text{mAP@0.5} + 0.9\,\text{mAP@0.5:0.95}$) on the validation split was selected. M2–M4 used the settings of Table II; the baseline M1 was trained with Ultralytics v8.0.141 and `optimizer=auto`, which selected AdamW ($\eta_0 = 0.002$, momentum 0.9). Default Ultralytics augmentation was applied: mosaic (disabled for the last 10 epochs), horizontal flip (0.5), scaling ($\pm$50%), translation ($\pm$10%), and HSV perturbation (0.015, 0.7, 0.4). Only single-run results are reported; run-to-run variability was not measured.

B. Evaluation Metrics

The detection performance was evaluated using Precision (P), Recall (R), Average Precision (AP), mAP@0.5, and mAP@0.5:0.95. AP is the area under the precision-recall curve; mAP@0.5 uses an intersection over union (IoU) threshold of 0.5, and mAP@0.5:0.95 averages IoU thresholds from 0.5 to 0.95. Metrics use Ultralytics [9] (v8.4.66; v8.0.141 for M1) with confidence threshold 0.001, NMS IoU 0.7, and at most 300 detections per image. Table III reports P and R at the maximum-F1 confidence threshold; Fig. 5(a) uses a separate fixed confidence threshold of 0.25 and IoU 0.45 for the confusion matrix.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Main Benchmark Comparison

Table III compares the baseline YOLOv8x (M1) with the proposed YOLOv8-CBAM (M4), and its last row lists the gain over the baseline. The proposed model improves all four evaluated metrics, with Precision rising by 4.48 and Recall by 4.85 percentage points.

The mAP gains indicate improved aggregate detection and localization on this validation split. Because only 6.8% of the validation boxes are small and the dataset contains no drone-free images, size-stratified AP and a negative-image evaluation were not conducted; small-object-specific gains and background false-alarm reduction therefore cannot be quantified. The near-duplicates found in Section III-A may also make these validation scores optimistic.

B. Ablation Study on Attention Components

To separate the effect of each attention branch, we compare four settings: the plain baseline (M1), channel attention only (M2), spatial attention only (M3), and full CBAM (M4), with each variant applied after Layers 12 and 15. Table III shows that both single branches improve on the baseline, with the spatial branch gaining a little more than the channel branch, while the full CBAM gives the best results on every metric, which shows that the two branches complement each other.

C. Computational Complexity Analysis

The two CBAM modules add 64,198 parameters: 51,299 at Layer 12 ($2 \cdot 640 \cdot 40$ MLP weights, 98 convolution weights, and γ) and 12,899 at Layer 15, raising the model from 68,153,571 to 68,217,769 parameters (+0.094%). Profiled directly at 640×640, GFLOPs rise from 258.1123 to 258.1141 (+0.0018, +0.0007%); fvcore gives the same increment, and

TABLE III
ABLATION STUDY OF ATTENTION CONFIGURATIONS

Configuration	Params	GFLOPs	P (%)	R (%)	mAP@0.5	mAP@0.5:0.95
M1: Baseline	68,153,571	258.1123	86.82	83.90	89.22%	53.76%
M2: Channel-Only	68,217,573	258.1126	88.54	85.12	90.71%	54.92%
M3: Spatial-Only	68,153,769	258.1139	89.15	86.40	91.25%	55.43%
M4: YOLOv8+CBAM	68,217,769	258.1141	91.30	88.75	92.93%	57.10%
Gain over Baseline (M4 − M1)	+64,198	+0.0018	+4.48 pp	+4.85 pp	+3.71 pp	+3.34 pp

Params and GFLOPs measured at 640×640 (GFLOPs $= 2\times$MACs). M1 trained with AdamW (Table II).

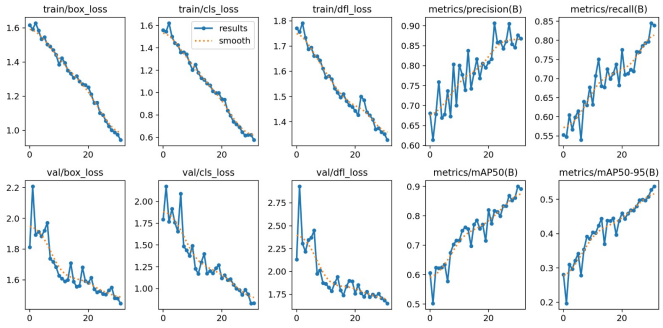
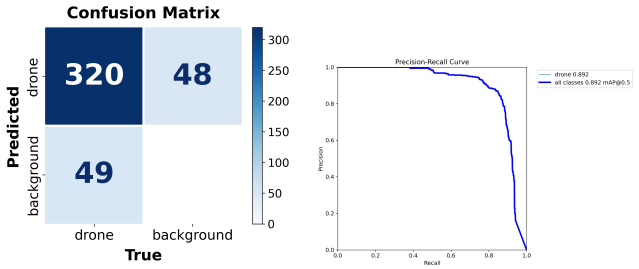


Fig. 4. Training loss curves and validation metric trajectories of YOLOv8-CBAM (M4) across 32 training epochs.



(a) Confusion matrix (b) Precision–Recall curve

Fig. 5. Validation results from different configurations: (a) confusion matrix of baseline YOLOv8x (M1), evaluated at confidence 0.25 and IoU 0.45; (b) precision–recall curve of YOLOv8-CBAM (M4). The panels represent different model configurations and operating points.

the uncounted element-wise attention operations add about 0.028 GFLOPs. The previously reported +0.1 GFLOPs came from Ultralytics' summary, which extrapolates from a 32×32 proxy image and inflates the size-independent MLP cost. On a 4-thread CPU (FP32, batch 1, interleaved runs), the median latency is 604.59 ms for the baseline and 606.91 ms for YOLOv8-CBAM (1.65 FPS each); the paired median difference of +7.67 ms (+1.27%) is within the run-to-run standard deviation of about 50–58 ms. GPU latency and memory were not measured, so real-time performance is not established.

VII. Visual and Qualitative Analysis

A. Training Convergence Dynamics

Fig. 4 illustrates the training convergence of YOLOv8-CBAM (M4) over 32 epochs. The losses decrease steadily without clear signs of overfitting, and the validation metrics stabilize during the final epochs.

B. Confusion Matrix and PR Curves

Fig. 5(a) shows the baseline (M1) confusion matrix, while Fig. 5(b) shows the proposed model (M4) PR curve; they represent different configurations. The matrix uses confidence 0.25 and IoU 0.45, whereas Table III reports precision and recall at maximum-F1 confidence. The displayed M1 matrix contains 320 true positives, 48 false positives, and 49 false negatives, corresponding to 86.96% precision and 86.72% recall at the matrix operating point. They differ from the M1 table values (86.82% and 83.90%) because the operating points differ. The M4 PR curve reports mAP@0.5 of 0.929. No negative-only split was evaluated.

C. Qualitative Detection Performance

Fig. 6(a) shows a representative detection produced by YOLOv8-CBAM for a close-range quadcopter captured against a clear sky, with out-of-focus palm foliage in the background. The model detects the drone with a confidence score of 0.79. No additional detection is visible on the textured vegetation in this example; this single image is not a quantitative background-rejection test. The predicted bounding box covers the main airframe and all four rotors, but it excludes part of the lower gimbal camera and the landing skids. Box boundaries can thus be imprecise for large near-field targets, consistent with the gap between mAP@0.5 and mAP@0.5:0.95.

D. Detection in Cluttered Ground-Level Scenes

Fig. 6(b) shows a multirotor drone above a tree line, with people and equipment in the foreground. The model localizes the drone (confidence 0.62); no additional detections are visible in this example, which does not quantify false-positive performance. The lower confidence than Fig. 6(a) accompanies the smaller apparent target and less distinctive thin structures.

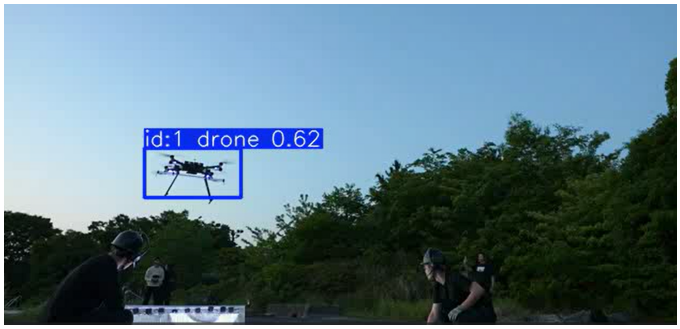
Observed failure conditions include severe motion blur, extreme distances where the drone occupies fewer than 8×8 pixels, and heavy occlusion behind buildings. Temporal multi-frame modeling may help address these cases.

VIII. Discussion

The results are consistent with the motivation for refining multi-scale features at the neck fusion points, where concatenated features may carry background information into higher-resolution maps. In the tested configuration, the spatial-only



(a) Close-range quadcopter against a clear sky



(b) Multirotor drone above a tree line in a ground-level scene

Fig. 6. Detection results of YOLOv8-CBAM: (a) a close-range quadcopter against a clear sky with blurred foliage in the background, and (b) a multirotor drone hovering above a tree line with people and equipment in the foreground. The overlay shows the track ID, class label, and confidence score; the track ID is from the Ultralytics tracker, used only for visualization.

variant scores slightly higher than the channel-only variant, while the combined CBAM variant obtains the highest reported validation scores. Because both attention branches were inserted at Layers 12 and 15 together, the current ablation does not isolate the accuracy contribution of either insertion point; their costs differ, with Layer 12 holding 51,299 of the added parameters (+0.0005 GFLOPs) and Layer 15 holding 12,899 (+0.0013 GFLOPs).

The evaluation is limited to one dataset and its supplied validation split. The duplicate audit shows that this split is not source-independent, and the dataset has few small targets and no drone-free images. Evaluation on the scene-grouped test split and on other datasets, size-stratified AP/recall, bird-like distractor and negative-image tests, comparison with drone-specific detectors, single-location and alternative-placement accuracy, comparison with another attention module, repeated-run statistics, and GPU profiling remain necessary. The reported gains should therefore be interpreted as single-run validation results rather than evidence of generalization or deployment readiness.

IX. Conclusion

This paper evaluated YOLOv8-CBAM, which applies channel and spatial attention after the 40×40 and 80×80 neck fusion stages. Pretrained weights initialize the model and are fine-tuned jointly with CBAM. On the supplied validation split, it reaches 92.93% mAP@0.5 and 57.10% mAP@0.5:0.95, gains of 3.71 and 3.34 percentage points, with 64,198 additional parameters (+0.094%) and 0.0018 additional GFLOPs. These single-split results do not establish independent generalization, small-object-specific gains, superiority, or real-time performance.

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